Response to Reviewer's Comments

Does Corruption Deter Female Leadership in Firms?
Manuscript JOIE-2025-0056

September 26, 2025

We thank the referees for re-reviewing the paper and for their comments, concerns, and suggestions. We also thank the editor for their feedback. In this document we detail our responses and the revisions that have been made to the paper. We hope that they are satisfactory and believe that the paper is now improved.

All quoted sections of the revised text that appear below are indented. (Since the revised text is "copied-and-pasted" into this document, please note that footnotes are numbered differently here than in the paper.) Editor and reviewer comments are in bold.

We begin with our response to the editor's suggestions.

1 Editor

EDITOR COMMENTS (partly in response to the report by referee 6 below)

The editors believe reviewer 6's main concerns are based on a misreading of the text. Furthermore, the IV analysis is a robustness check and not the main empirical presentation. The authors are careful to not overstate causality.

IV Approach: Instruments (1998 IQIM data on councils existence/activity, management capacity; judiciary district presence) are predetermined, reducing reverse causality. First-stage F-stats are strong. J-stats often fail to reject exogeneity (though not always, and the test has low power). Just-identified results (Appendix E) show consistency. Paper is cautious, treating IV as robustness only, not claiming strong causality.

The sectoral argument is a misreading of the paper. For example, the paper notes low female participation in corrupt sectors (2.1% vs. 33% overall), but effects are on leadership shares conditional on participation. Reviewer 6 misinterprets as "moderate drop" but leadership share rises from 6.36% workforce share to 20%, suggesting over-representation in leadership within those sectors. But the paper focuses on corruption's marginal effect.

Lack of time variation is because of using subnational municipal audit data. Transparency International, for example, is only at national level.

Sample Selection: The paper is based on Avis et al. dataset for consistency. Appendix C suggests no major differences.

IVs/First Stage: This is a misreading of the instruments. Informal economy is a control, not IV (in "included instruments," standard in 2SLS). Judiciary district may correlate with size/audit probability, but audits are random.

There is always a concern regarding the exclusion restriction. The editors do not think the authors oversell the IV results, thus are fine with them remaining in the paper. But, we simply ask for the IV section to be removed or condensed since it is a robustness check.

Author Response: We greatly appreciate the clarity in this guidance, particularly in responding to Reviewer 6. We have removed the instrumental variable results entirely from the main text of paper and have placed them all in Online Appendix E. We only note that they exist in foonote 8, page 9. This footnote copied below.

> "Our focus is on the OLS results and we are careful not to make overly causal statements in the paper. However, we also pursue an instrumental variable analysis via Two-Stage Least Squares (2SLS) in Appendix E. We explain and motivate the external instruments used in the Appendix."

2 Reviewer 1

Comments to the Author

The authors examine the impact of corruption on the propensity of women to attain leadership positions in Brazil, using both cross-sectional methods and IV analysis. Their analysis is extremely thorough, with various different specifications responding to concerns I had with their paper when reading through the OLS section. It is clear that they put some significant thought into their analysis, and have a very comprehensive empirical strategy. I have only a few minor comments: You mention California's gender quota law in footnote 2. It is currently not in force (suspended pending appeal). You keep saying there are 935 municipalities that were audited/you can use in your study, but you never mention how many municipalities there are in Brazil total. It would also be helpful to briefly mention how these municipalities compare to the US - sort of like counties, but also sort of like cities. Really, what I'm looking for is a defense of why examining this question at the municipality level is appropriate, beyond just saying that it is at this level that you have data.

Author Response: We appreciate the thoughtful comments and are happy you agree that the paper has improved. We respond to each new (minor) comment in turn.

We have revised the footnote to clarify that California's law is no longer in force.

In detailing the data, we do state the total number of municipalities in Brazil and carefully explained how/why we are left with 935 in our study. For example, this is text from page 10:

> "Brazil has a total of 5,570 municipalities. From 2003 through 2015, 1,949 of these municipalities experienced at least one audit with 2,241 audits in total (Avis, Ferraz, and Finan 2018)."

And, after a page of discussions on the sample, we write this (page 11):

> "This leaves with us 935 municipalities for which we can study the corruption–female leadership relationship."

But, to clarify further we have edited page 3 to read:

> "We test whether higher levels of corruption impact female leadership across Brazilian municipalities. Our corruption data is derived from the public release of reports concerning results from random audits of municipal expenditures and includes 935 (of 5,570) municipalities."

We have also added some additional discussion of municipalities in Brazil and why they are of interest for our study. See page 3, copied below.

> "We investigate whether higher corruption levels affect female leadership in Brazilian municipalities. These municipalities exhibit remarkable diversity, with GDP per capita ranging from R$2,261 to R$298,919, urbanization rates spanning less than 1% to at most 99%, and informality accounting for 13 to 97% of employment. Corruption also varies significantly, with Ferraz and Finan (2011) estimating that the share of resources tied to corrupt activities ranges from 0% to nearly 80%. This substantial variation in both corruption and economic indicators makes Brazil an ideal setting for studying our research question."

3 Reviewer 2

Comments to the Author

The authors have done a great job addressing the issues raised in the previous review. I have no further concerns.

Author Response: Thank you.

4 Reviewer 3

None.

5 Reviewer 4

Comments to the Author (There are no comments.)

6 Reviewer 5

None.

7 Reviewer 6

Comments to the Author

Referee report

Does Corruption Deter Female Leadership in Firms?

The authors aim to analyze the effect of corruption on female leadership in Brazil. Based on cross-sectional municipal data, they conclude that corruption is the primary factor deterring women from pursuing managerial careers. I found the topic of the paper interesting and relevant; however, the identification of the effects is not done adequately. Furthermore, the various biases in data sampling are not addressed, and the use of the instruments is also problematic.

Main Comments:

1) The weakest point in the analysis is the identification of the links between corruption measures and female participation. The sample selection (data availability and different intensity of audits) is not addressed. And also, the direction of causality. Are women leaving (not entering) leadership roles in the corrupted regions, or is corruption more widespread in areas with low female participation? Also, is it female labor participation or, more precisely, female participation in politics? Let me cite the authors on how they argue the "stronger" links in "corrupt" industries: "Only 2.1% of working age women are employed in these sectors, though 33% of women in this group are working in general (across any sector). Thus, these industries comprise only 6.36% of total female workers in the economy; this is compared to the 19.72% of total employment that these industries constitute overall. Perhaps more importantly, women are less likely to be leaders within these industries. Female leaders as a share of total leadership positions across all industries averages 33.1%, whereas female leaders within these four sectors is less than 20%."

This line of argument is standard throughout the paper. In one part, the authors acknowledge and cite that female labor participation varies significantly across different industries (here, 2.1% versus 33%) and then argue that this disparity translates into much lower female leadership participation (from 33.1% to 20%). A general reader would interpret that these industries are rather specific in terms of demand for technical education, manpower, or other skills. A dramatic difference in female labor participation is transferred to a relatively moderate drop in the leadership share. (In other industries, the share in labor participation is the same as the leadership; in "problematic" industries, shares increased from 2.1% to 20%)

Author Response: Thank you for these comments. First, to address the quoted section, we want to be clear that this is simply a discussion of the summary statistics and is not intended to make a claim about the marginal effect of municipal corruption. We have added a sentence at the end of the quoted paragraph to clarify (copied below).

> "However, they are unconditional averages and differences should not be interpreted as a marginal effect of municipal corruption. We must estimate this effect using OLS regressions, and do so in the following section."

Second, it is unclear why the reviewer is comparing statistics that have different denominators. The share of female leaders (20%) is an indicator that is interpreted relative to men. The share of working women that are working in corrupt sectors (2.1%) is not. Yes, the drop in this latter share relative to all sectors (33% versus 2.1%) is bigger in magnitude than the drop in the former (33% to 20%), but it is unclear why one would expect these changes to be the same. In fact, self-selection theory could imply that the women that are remaining in the corrupt industries are *more* likely to be leaders. We discuss this when motivating why one needs to look at both sets of outcomes (relative to men and participatory outcomes) to understand the full picture. This is a quote from page 4 where this is explained.

> "We believe both sets of outcomes are necessary to understand the nature of the corruption–female leadership relationship. For example, if corruption is a deterrent to working women, we may find that it reduces female labor force participation. If this effect is specific to women (i.e., does not impact male labor force participation), we should see a decrease in the share of leadership positions held by women. However, if corruption also induces a selection effect such that the women that remain in the labor force are the ones that are more likely to hold positions of power, corruption's effect female presence among leadership (relative to men) could be positive. Alternatively, if the selection effect discourages working women from obtaining positions of leadership, we might see all outcomes moving in the same direction. These are all empirical questions."

We don't discuss this explicitly with regard to these summary stats because our goal is to estimate municipal corruption's marginal effect on leadership and participation. This relative change across industries, while interesting, is not the primary focus of our paper. We just use these statistics to motivate the idea that things are different in these industries.

As for the other comments in this section, we want to be clear that we are careful in not making causal claims. And, this is a paper about female labor participation. We say nothing about female participation in politics.

2) Sample selection bias.

Let me cite the authors: "Brazil has a total of 5,570 municipalities. From 2003 through 2015, 1,949 of these municipalities experienced at least one audit with 2,241 audits in total (Avis et al., 2018). Because Avis et al. (2018) focus on two electoral terms (namely, 2004-2008 and 2008-2012) and because audits ceased the random component in 2015, their dataset focuses on corruption uncovered in audits that occurred anytime between July of 2006 to March of 2013; i.e., lotteries 22 through 38."

On the other place they said: "Our cross-sectional sample is limited to the 935 municipalities that were audited and have corruption data available." However, if the corruption is estimated based on the audit data, why do they not construct measures for the remaining 1,000 municipalities? Also, what is the relationship between being audited and estimated corruption?

If it is random, then the different audit circles should have about the same distribution. The fact that some municipalities are audited more times should not be statistically different.

Author Response: We believe that we are already clear in our description of the data. Our data comes from Avis, Ferraz, and Finan (2018); they construct the corruption measures in a careful and consistent manner. We use their data as a baseline as it was produced for and published in a top-tier economics journal (*Journal of Political Economy*). However, there are other measures that we could use from other works, though they are more limited in terms of municipal coverage. These other works are also published in top-tier economics journals (e.g., *American Economic Review*). We produce these results in the appendix and the results are unchanged.

The audits are intended to reduce corruption. While they are random and should not be influenced by corruption itself, the entire thesis of the Avis et al. JPE paper is that the audits reduce corruption and we thus should see a lower corruption level in subsequent audits.

3) The authors assume that corruption is persistent and not changing. I agree that corruption is persistent, possibly changing slowly, but it is changing. No time variation (?), what about using Transparency International data for global trends in Brazil https://www.transparency.org/en/countries/brazil

Author Response: It is the outcome variables that are limiting the time variation as they are derived from the Census. The corruption data does have a degree of time variation, though this variation is often *not* within-municipal variation (only some are audited multiple times). Using country level trends from something like Transparency International would not be beneficial for several reasons. First, adding this time variation would eliminate all municipal variation – and given the diversity of Brazil – we believe this is a major drawback. Second, we would be using an imprecise indicator of corruption that relies on perceptions and not facts. Third, our outcome variable is still limited to decennial Census data.

4) Use of instruments and the first stage regression. Some measures used as instruments (e.g., an indicator of a juridical district) may be associated with municipality size, as well as with the number of economically active individuals and, therefore, the probability of being audited. The use of IV estimation is not correct (as seen in Appendix E). The percentage of informal activities is not a valid instrument; shadow economy activities are usually strongly correlated with corruption. They even mentioned it on page 16, but then included informality between the instruments. This is what strikes me: the authors admit or describe certain phenomena and link them to the existing literature in certain parts of the text, but then ignore this in their methodology and estimation a few pages later.

Author Response: We apologize for the abruptness of this response but this is an incorrect assessment of (1) what we do and (2) how 2SLS is done more generally. We are clear that this IV analysis should just be used as a robustness check and have moved it to an appendix entirely. But, more importantly, we do not use the size of the informal economy as an *external* instrument. In the first stage of all IV estimations, one uses both external IVs and all control variables as instruments. This does two things: helps ensure conditional exogeneity and avoids omitted variable bias in the first stage. This is standard practice and how IV analyses are conducted.

So minor comments

5) What is the spill-over effect of neighboring regions?

Author Response: It is beyond the scope of this paper to study this here but we did include a paragraph in the concluding section discussing spillovers and the need for future research. (See page 21.)

6) What is the role of audits? Aren't they related to tax collection or the use of public funds? It is unclear.

Author Response: The role of the audits is to reduce corruption in the use of federal transfers. This is discussed in Section 3.2 on page 10. Here is a quote from this section.

> "The goal of the audits is to uncover malfeasance in the use of public funds."

7) Is there any information about the size and ownership type of the firms used to comprise the sample? It also affects female participation.

Author Response: Unfortunately, we do not have this information as we use decennial Census data, which covers households and not firms.

8) The authors say on page 4: "The second set of outcomes examine only the female labor force and the types of jobs that women hold. That is, this second set of outcomes should not be interpreted relative to men. Corruption may reduce the availability of jobs and leadership positions in particular, regardless of gender. This is still an important deterrent to consider." It is unclear how corruption generally affects the availability of leadership positions studied by the authors. I presume that the district's composition is not changed (much) and the number of political representatives is not reducing due to corruption and is relatively stable. If not, then the authors should consider this factor in their analysis as well. Otherwise, it is always relative to men.

Author Response: This is not a correct interpretation. Corruption can certainly impact sectoral composition and we discuss this in the conclusion. Regardless, this measure is entirely relative to women – men do not appear in these measures in any way. If corruption reduces leadership positions in a certain area (which we do not *know* because we do not estimate this) and we see a reduction in female leaders as a percentage of female workers, we could see a larger, smaller, or equal change for men.